

DIFFUSION MODELS FOR SCORE-BASED UNCERTAINTY QUANTIFICATION

Anonymous authors

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ABSTRACT

(plus haut niveau mais technique)

We propose a post-hoc density-based method for out-of-distribution (OOD) detection built on the score-based formulation of diffusion models. Rather than relying on a computationally costly reconstruction error or ad-hoc method that use the score as a proxy for density, our approach integrates the learned score function of a diffusion model along a path between the point of interest and a reference point to approximate the true density $\log p(x)$ up to a constant. Since diffusion model score error compared to the true score is greater the further away we are from the training data density, it is crucial to choose a reference point x_{ref} in a high-density region to keep the path short in low-data zones and reduce errors that arise. To achieve this, we perform gradient ascent on the learned score to locate a suitable x_{ref} . Experiments on both 2D synthetic data (Two Moons) and real-world image embeddings (CIFAR-10 vs. CIFAR-100) show that our method outperforms reconstruction-based and other score-based method in the literature. We conclude by highlighting potential extensions to higher-dimensional domains and discussing how this technique could be used with large-scale diffusion models trained once for multiple tasks.

1 INTRODUCTION

OOD detection, also known as anomaly or outlier detection, aims to detect abnormal samples that lie far from a given data distribution. It is crucial in high-stakes fields like robotics, healthcare, and finance, where robust uncertainty quantification (UQ) can prevent risky decisions. A practical OOD detector should remain post-hoc, relying only on in-distribution (ID) data without assuming prior knowledge of outliers, which cannot be exhaustively defined. Recently, diffusion models (Song et al., 2021; Ho et al., 2020) have gained attention for their capacity to sample complex data distributions, making them attractive for OOD detection.

1.1 OOD DETECTION METHOD USING DIFFUSION MODEL

Existing OOD detection methods that employ diffusion models can be classified into three types:

Reconstruction based method: This type of approach corrupts the point of interest, reconstructs it via a diffusion model, and uses the distance between the original and reconstructed points as a proxy for density. One of the first methods to apply this intuition is LMD (Liu et al., 2023), which raises three problems.

First, the metric used here is ad hoc: the distance between reconstructed and original data is assumed to correlate with density, but no theoretical background guarantees this. Second, there is the question of how to corrupt the data. One could argue in favor of adding Gaussian noise, since diffusion models are trained by learning to predict Gaussian noise added to images. However, LMD instead uses masking to corrupt part of the data, and there is no guarantee that a diffusion model trained on Gaussian noise can reconstruct data altered by masking. (Yao et al., 2025) addresses this issue by introducing synthetic abnormal samples in training, but that requires a specialized diffusion model training procedure and thus reduces the applicability to pretrained models.

Third, there is the question of how much information should be deleted/corrupted for the method to work. It ultimately depends on how distant the in-distribution and out-of-distribution data are from each other; ideally, we should not have to assume anything about OOD data for broader applicability. (Bercea et al., 2023) deals with this by integrating automatic masking, stitching, and re-sampling techniques for anomaly detection.

Score-based method Score-based methods leverage the fact that diffusion models learn the score (i.e., gradient of the log density) at different timesteps, and use it as a metric for OOD detection. (Mahmood et al., 2020) proposes using the norm of the score at each timestep as a vector to quantify uncertainty, then fitting a Gaussian kernel on this vector space using ID data. This is promising because a low score could indicate high data likelihood. However, the score only provides the gradient of the log density and gives far less information than the density itself. For example, the gradient of the log density may be small in regions where the curvature of the density changes, and such regions could be incorrectly treated as in-distribution. To alleviate this, (Heng et al., 2024) proposes examining higher derivatives of the log density by computing the gradient of the score (the Hessian). This only partially resolves the issue, as it merely considers a higher-order Taylor expansion, and regions where the first n derivatives vanish can still occur. Nevertheless, score-based methods remain promising because (Han et al., 2024) offers theoretical assurances that diffusion models approximate the score well, also providing an upper bound on the approximation error.

Score integration We propose a novel UQ method that can be used for OOD detection. By integrating the score along a path between x and y in latent space, we can theoretically recover the exact log density up to a constant. This method does not suffer from the lack of theoretical grounding seen in reconstruction-based methods, since we are (in principle) recovering the density itself. In practice, the density is an approximation because the score is learned via a trained diffusion model, but (Han et al., 2024) shows that the approximation error is bounded. Thus, we can derive a corresponding bound on our integrated log density. Unlike (Mahmood et al., 2020; Heng et al., 2024), our approach does not rely on first- or second-order Taylor approximations, and it can distinguish a region where the first n derivatives of the density vanish from a true in-distribution region by integrating between x and a high-density reference point x_{ref} .

We summarize our contributions as follows:

Contributions.

- **Theoretical grounding:** We propose a novel *score integration* method for diffusion-based OOD detection which inherits a clear theoretical foundation via the Gradient Theorem, ensuring an exact log-density.
- **Broader scope of application:** This method does not suffer from Taylor approximations like other score-based methods do, and does not need to be tailored for specific tasks as reconstruction-based methods do, making it applicable in a wider range of use cases.
- **Unified error bound:** We build on the theoretical results of (Han et al., 2024), which bound the approximation error of a learned score, to show that our integrated log-density similarly benefits from a known upper bound on its accuracy.
- **Experiences:** We conduct experiments on near-OOD CIFAR-10 vs. CIFAR-100 and demonstrate that our method surpasses other diffusion-based methods for OOD detection.

2 BACKGROUND

2.1 SCORE-BASED DIFFUSION MODELS

The diffusion model forward process is defined by the following SDE.

$$dx = f(x, t) dt + g(t) dW_t$$

where $f(x, t)$ is the drift term, $g(t)$ is the diffusion coefficient controlling the intensity of the noise, dW_t is a Wiener process, and x_0 follow the data distribution. In the case of the variance-preserving formulation, the forward process becomes

$$dx_t = -\frac{1}{2}g(t)x_t dt + \sqrt{g(t)} dW_t$$

We denote by $p_t(x)$ the probability density of x_t . We can define the reverse of this SDE, starting from an initial Gaussian distribution. The resulting distribution $\tilde{x}_T$ of the reverse process, for a sufficiently large T , matches that of the forward process, which is the distribution of training data. Anderson (1982); Haussmann & Pardoux (1986).

$$d\tilde{x}_t = \left(\frac{1}{2}\tilde{x}_t + g(t)\nabla_x \log p_{T-t}(\tilde{x}) \right) dt + d\tilde{W}_t$$

We can then sample a new data point by solving this reverse SDE. In order to do so, we need to train a model to estimate the logarithmic likelihood gradient $\nabla_x \log p_t(x)$. Multiple loss have been proposed, but they can all be interpreted as a special case of a unified formulation proposed by Diederik P. Kingma (2023).

We will introduce the formulation proposed in this paper. We will define λ_t as the amount of noise, which is the log SNR, added by the forward process at the time step t , we have the following relation $g(t) = \frac{d}{dt}(1 + e^{-\lambda_t})$. The meaning of λ_t can be understood through the discretisation of the forward process, which give us

$$x_t = \sigma(\lambda_t)x + \sigma(-\lambda_t)\epsilon$$

σ being the sigmoid function. We then have $\lambda_t = \log \frac{\sigma(\lambda_t)}{\sigma(-\lambda_t)}$, which is the log signal to noise ration. The unified loss proposed is below.

$$L(\theta) = \frac{1}{2} \mathbb{E}_{x_t \sim p_t, \epsilon \sim \mathcal{N}[0, I], \lambda \sim \mathcal{P}(\lambda)} \left[\frac{\omega(\lambda)}{p(\lambda)} \|\epsilon - \epsilon_\theta(x_t, t)\|^2 \right]$$

The function mapping the diffusion step t to the log-SNR λ is called the noise schedule and we have $g(t) = -\frac{d}{dt} \log(\bar{\alpha}_t)$ and $\lambda = \log(\frac{\alpha_t}{1-\bar{\alpha}_t})$. ω is a weighting function and $\mathcal{P}(\lambda) = -\frac{d\lambda}{dt}$ is the distribution of noise intensities. The trained model will then output $\epsilon_\theta(x_t, t) \approx -\sigma_t \nabla_x \log(p_t(x_t))$.

Where $\sigma_t = \sqrt{1 - \bar{\alpha}_t}$ and $\bar{\alpha}_t = \exp\left(-\int_0^t g(s) ds\right)$

As stated in Diederik P. Kingma (2023), the noise schedule acts as an importance sampling for the diffusion step, and if the weight function is monotonic, it is equivalent to a data augmentation method with additive noise.

We will note $s_\theta(x, t) = \frac{\epsilon_\theta(x, t)}{-\sigma_t} \approx \nabla_x \log(p_t(x))$ for the rest of the paper.

The most common noise schedule being use in practice is the cosine schedule Nichol & Dhariwal (2021), which have been show to help generate sample of better quality.

experiment We will do experiment on synthetic data set in low dimenssion to highlight some properties of diffusion model, we will then apply our method on image, specifically cifar-10 as in distribution and cifar-100 as out distribution. For images, we will always work in a syntactic latent space.

3 PROPOSED METHOD

3.1 SCORE INTEGRATION

Since diffusion model learn the score at multiple diffusion step, in particular diffusion model give us access to the score of the training data set $s_\theta(x, 0) = \nabla_x \log(p(x))$. Our method then builds on the fundamental relationship between score functions and density estimation through the Gradient Theorem. For any smooth path $\gamma : [0, 1] \rightarrow \mathbb{R}^d$ connecting a reference point x_{ref} to a query point x_0 , we have:

$$\log p(x_0) - \log p(x_{\text{ref}}) = \int_\gamma \nabla_x \log p(x) \cdot dx \quad (1)$$

When using the true score function $\nabla_x \log p_t(x)$, this equality holds exactly. For learned scores $s_\theta(x, t)$ via diffusion model, the approximation error becomes:

$$\left| \int_{\gamma} (s_\theta(x, 0) - \nabla_x \log p(x)) \cdot dx \right| < \text{length}(\gamma) \cdot \epsilon_{\text{score}} \quad (2)$$

where ϵ_{score} is the maximum pointwise error. This motivates keeping integration paths short in high-density regions. We will choose, for simplicity, a straight line path $\gamma = tx_{\text{ref}} + (1-t)x_0, t \in [0, 1]$. After parametrizing the integral we have:

$$\int_{\gamma} \nabla_x \log p(x) \cdot = \int_0^1 \langle \nabla_x \log p(tx_{\text{ref}} + (1-t)x_0), \frac{d\gamma}{dt}(t) \rangle dt \quad (3)$$

Using Gauss-Legendre quadrature with n points, we approximate:

$$f(x_0) \approx \frac{1}{2} \sum_{i=1}^n w_i s_\theta(z_i, 0) \cdot (x_{\text{ref}} - x_0) \quad (4)$$

Where $z_i = x_{\text{ref}} + t_i(x_0 - x_{\text{ref}})$ and $\{t_i, w_i\}$ are quadrature nodes/weights. In our proposed method, evaluating the function s_θ can be done by a forward pass of a neural network. This can be costly, the choice of using Gauss-Legendre quadrature arise from the fact that this method approximate an integral very well with few evaluating point, when the function to integrate can be approximated by a polynomial function. It can be showed Gauss (1814) that Gauss-Legendre quadrature is optimal in the sense that for n node it exactly integrates a polynomial of degree $2n - 1$ or less. It can be shown that no other quadrature rule with n nodes can do this or better. Since neural network are piecewise linear function, they can be well approximated by a polynomial function. There is effective algorithm to compute quadrature node and weight for large n , since we found that in practice our proposed method never need more then a 100 quadrature node, the computational cost of computing quadrature node and weight is negligible.

3.2 REFERENCE POINT SELECTION

The choice of reference points critically impacts approximation quality. In order to keep integration path short and in high density region, we will initialize the reference point μ^0 by the mean of the training data set, and we will then perform a gradient ascent algorithm on the learned score $s_\theta(x, 0) \approx \nabla_x \log(p(x))$. This will result in a reference point close to the center of the data set, and in a local maxima of the log density.

3.3 DENSE COVERAGE SCORE INTEGRATION

The score integration method proposed in the previous section work in practice but it can be improve, the pseudocode of this method can be found in the appendix. This method is sensitive to the reference point selection, point further away from the reference point will have an undervalued log density because of the high score error the diffusion model make in low density region, furthermore a reference point stuck in a local maxima of low density can reduce the effectiveness of our method. To solve this problem, we first perform a k-means on training data set, each centroids μ^j will be the initialization of a gradient ascent algorithm. This will result in the selection of k reference point $(x_{\text{ref}}^j)_{1 \leq j \leq k}$. This ensure a dense coverage of the training data set space, sample from the training data distribution will be more likely to be close to at least one reference point, the log density difference of at least one reference point will not be undervalued. This will define a feature space of k log density difference between the training data and the reference point $(f_j(x))_{x \in X_{\text{train}}, 1 \leq j \leq k}$. We can then train a KDE on this feature space, the density estimation of this KDE will be our uncertainty quantification value.

Algorithm 1 Training Phase: Reference Points & KDE Estimation

Require: ID training data X_{train} , diffusion model s_θ trained on X_{train} , number of quadrature nodes n , quadrature nodes $(t_i)_{1 \leq i \leq n}$, quadrature weight $(\omega_i)_{1 \leq i \leq n}$, number of cluster k

Ensure: Reference points $\{x_{ref}^j\}_{j=1}^k$, KDE model g_{KDE}

- 1: Cluster X_{train} into k groups via K-means $\rightarrow \{\mu_j\}_{j=1}^k$
 - 2: Refine each μ_j via gradient ascent:
 - 3: $\mu_{n+1}^j \leftarrow \mu_n^j + \alpha_n s_\theta(\mu_n^j, 0)$
 - 4: $(x_{ref}^j)_{1 \leq j \leq k} \leftarrow (\mu_N^j)_{1 \leq j \leq k}$
 - 5: For each $x \in X_{train}$, compute each log density features:
 - 6: $f_j(x) = \frac{1}{2} \sum_{i=1}^n \omega_i s_\theta(z_i, 0) \cdot (x - x_{ref}^j)$ where $z_i = x_{ref}^j + t_i(x - x_{ref}^j)$
 - 7: Train KDE g_{KDE} on $(f_j(x))_{1 \leq j \leq k, x \in X_{train}}$
 - 8: **return** $(x_{ref}^j)_{1 \leq j \leq k}, g_{KDE}$
-

Algorithm 2 Inference Phase: OOD Scoring

Require: Test sample X_{test} , reference points $(x_{ref}^j)_{1 \leq j \leq k}$, KDE g_{KDE} , diffusion model s_θ trained on X_{train} , number of quadrature nodes n , quadrature nodes $(t_i)_{1 \leq i \leq n}$, quadrature weight $(\omega_i)_{1 \leq i \leq n}$

Ensure: OOD score $s_{ood} \in \mathbb{R}$

- 1: For each $1 \leq j \leq k$ and test sample $x \in X_{test}$:
 - 2: $f_j(x) = \frac{1}{2} \sum_{i=1}^n \omega_i s_\theta(z_i, 0) \cdot (x - x_{ref}^j)$ where $z_i = x_{ref}^j + t_i(x - x_{ref}^j)$
 - 3: Compute density estimate: $s_{ood} = -\log g_{KDE}((f_j(x))_{1 \leq j \leq k, x \in X_{test}})$
 - 4: **return** s_{ood}
-

3.4 ON THE IMPACT OF DIFFUSION MODEL LOSS

Our proposed method suppose that the diffusion model approximate well $s_\theta(x, 0) \approx \nabla_x \log(p(x))$. But as shown by Kingma et al. (2021), noise schedule have an impact on the loss of our diffusion model, the diffusion model loss stated in the background is the Monte Carlo estimator of the following integral

$$\int_{\lambda_{min}}^{\lambda_{max}} \omega(\lambda) E_{\epsilon \sim \mathcal{N}[0, I], x \sim p} (||\epsilon - \epsilon_\theta(x_\lambda, \lambda)||^2) d\lambda$$

With $\lambda_{max} = f_\lambda(0)$ and $\lambda_{min} = f_\lambda(1)$ being the endpoint of the noise schedule function. If $t_{min} > 0$, then the diffusion model will never be trained for low noise level $t = 0$. For our proposed method to work we either need to ensure our diffusion model is trained on an noise schedule function that visit low noise level with $t_{min} \leq 0$, or we need to integrate $s_\theta(x, t_{min})$. This score don't exactly correspond to the training data score, but the resulting value is typically close to the training data score since λ_{min} is usually low enough to ensure good sample quality.

We could also try to improve our methods by tailoring a noise schedule that increase the precision of the approximation $s_\theta(x, 0) \approx \nabla_x \log(p(x))$ made by our diffusion model. Since noise schedule work as importance sampling, we could do so by choosing a probability distribution of the log SNR λ with high value, corresponding to a higher probability to sample around $t = 0$ for the Monte Carlo loss.

The weighted loss could also be tailor to improve the score estimation of the diffusion model, by using a high weight for low noise intensity.

3.5 VISUALIZING ON TOY DATA SET

To check the coherence of using a diffusion model to approximate data distribution score, we can check on a toy data set, two moon data set, the score estimated by the diffusion model and the true score obtain via KDE estimation.

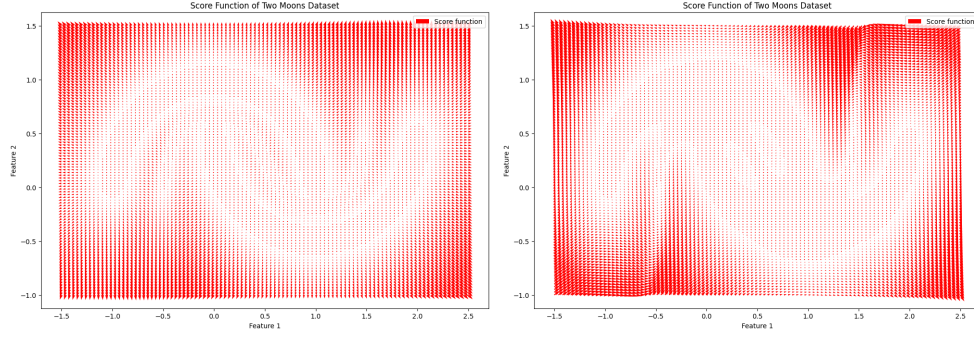


Figure 1: Left: Vector field of the score estimated by the diffusion mode Right: Vector field of the true score estimated from the KDE density

We can see that the score estimation seems good for in distribution region, but the further away we are from the data distribution, the worse the diffusion model score approximation become.

4 ABLATION

5 RELATED WORK

We review how diffusion models (Song et al., 2021; Ho et al., 2020) have been employed in out-of-distribution (OOD) detection and uncertainty quantification (UQ). We focus on two main categories of diffusion-based OOD methods: *reconstruction-based* and *score-based*. We also briefly discuss a *hybrid* approach that combines both reconstruction and score information.

Reconstruction-based Methods. Reconstruction-based approaches typically corrupt the input data and then attempt to restore it using a generative model trained only on in-distribution samples, under the assumption that out-of-distribution points will be harder to reconstruct. One of the earliest diffusion-based examples is LMD (Liu et al., 2023), which masks part of the image and uses a diffusion in-painting process to estimate how well the masked portion can be recovered.

Despite their empirical success, reconstruction-based methods lack a solid theoretical grounding. They rely on the heuristic that reconstructed sample that end up far from the original sample indicate OOD samples, even when the chosen corruption process don't aligns with the training procedure of the diffusion model (Gaussian noise). Yao et al. (2025) proposes adding synthetic anomalies during training to ensure that a diffusion model can handle the corruptions use for OOD detection, but this requires specialized training and reduces the utility of off-the-shelf pretrained models. Another challenge is choosing how much to corrupt the sample, this can vary based on how distant the OOD data are from the in-distribution data, but since we don't have access to OOD information when we do unsupervised OOD detection, this hyper-parameter is hard to set and no procedure can help us. Bercea et al. (2023) tackles this issue using adaptive masking and stitching techniques. Finally, obtaining a reliable reconstruction can require many forward passes, this make reconstruction-based methods computationally expensive.

Score-based Methods. Score-based methods leverage the fact that diffusion models estimate the score function, i.e., the gradient of the log-density. Intuitively, if the score is high or low in certain regions, it may indicate whether the sample is near or far from high-density regions of the training distribution. A prominent example is MSMA Mahmood et al. (2020), which evaluates the score at various time steps and uses the resulting score vectors to detect anomalies. However, a key drawback of score-based methods is that they only provide local gradient information. In regions where higher-order derivatives vanish, relying solely on the first-order score can lead to misclassification.

To mitigate these limitations, Heng et al. (2024) proposes examining the Hessian (the derivative of the score), allowing the detection of curvature changes in the log-density. While this partially addresses the issue, it remains reliant on higher-order Taylor approximations, certain pathological

OOD regions might still appear in-distribution if enough derivatives vanish. Score-based Methods are still promising because it have a better theoretical grounding then reconstruction-based methods. Indeed, the score is directly link to the density, and Han et al. (2024) establishes theoretical guarantees that diffusion models approximate the score well and provide an upper bound on the approximation error—an encouraging sign for the viability of score-based methods overall.

Hybrid Approaches. Beyond purely reconstruction-based or score-based methods, a few hybrid approaches use both reconstructed samples and their score signals. For example, Wang et al. (2023) applies a diffusion-based reconstruction step and then evaluates the score of the reconstructed sample. This strategy can improve detection accuracy but also compounds the computational cost and inherits theoretical gaps from both reconstruction- and score-based paradigms.

In summary, existing diffusion-based OOD detection methods either require expensive reconstructions for robust performance, rely solely on local gradient information, or combine both. Our work addresses these limitations by proposing a method that integrates the score along a path in a latent space, yielding a more direct log-density estimation, offering a clearer theoretical foundation via the Gradient Theorem. This method also don’t suffer from the expensive cost reconstruction error and NLL have.

6 EXPERIMENTS

6.1 EXPERIMENTAL SETUP

We describe the datasets, Two Moons data set vs background point selected on a grind, and CIFAR-10 vs CIFAR-100 projected on the latent space of a classifier trained on CIFAR-10. We then discuss the architectures for our diffusion-based score networks, MLP for 2D, 1D U-Net for higher dimensional embeddings. We also specify standard metrics like AUROC for OOD detection, and outline the baselines we compare against,

Table 1: Comparison of Different OOD Detection Methods.

Method	Approach	Uses Diffusion?	Post-hoc?
Diffpath 6D	Classification-based	Yes	Yes
DDPM-OOD	Reconstruction-based	Yes	Yes
MSMA	Classification-based	Yes	Yes
Ours	Density-based	Yes	Yes

6.2 PRELIMINARY EXPERIMENTS AND ABLATION STUDIES

We explore how the number of quadrature nodes impacts accuracy and time complexity, how different x_{ref} choices affect performance, and how gradient ascent can yield a stable high-density reference point.

6.3 TWO MOONS: SYNTHETIC ANALYSIS

We first train our diffusion model to sample correctly two moon dataset.

On the Two Moons dataset, our line-integration method yields high AUROC in detecting OOD background points, outperforming existing score based method or reconstruction-based method.

6.4 EXPERIMENTS: LATENT-SPACE DIFFUSION

Here, we apply our method to ResNet50 embeddings of CIFAR images. Using a 1D U-Net architecture for diffusion in the embedding space, we compare OOD detection vs CIFAR-100. Again, explicit log-density estimation via score integration proves effective, showing better AUROC than existing score based method and reconstruction-based approaches.

Table 2: OpenOOD results for CIFAR-10 and CIFAR-100. All values are AUROC; NFE is the number of function evaluations. The **Mean** column averages the six OOD datasets for each in-distribution dataset.

Method	CIFAR-10								CIFAR-100							
	Near-OOD			Far-OOD			Mean	NFE	Near-OOD			Far-OOD			Mean	NFE
	CIFAR-100	TinyImgNet	MNIST	SVHN	Texture	Places365	Mean		CIFAR-10	TinyImgNet	MNIST	SVHN	Texture	Places365	Mean	
Diffusion-Based Methods																
Diffpath 6D	0.840	0.857	0.948	0.889	0.871	0.867	0.879	20	0.592	0.634	0.608	0.686	0.606	0.582	0.618	-
NLL	0.754	0.788	0.946	0.823	0.715	0.874	0.817	1000	0.737	0.746	0.707	0.754	0.674	0.726	0.724	-
MSMA	0.603	0.670	0.756	0.636	0.726	0.656	0.674	10	0.608	0.529	0.488	0.533	0.639	0.513	0.552	-
DDPM-OOD	0.886	0.908	0.939	0.921	0.927	0.905	0.914	250	0.670	0.739	0.708	0.694	0.728	0.685	0.704	-
Non-Diffusion Methods																
ODIN	0.796	0.810	0.966	0.845	0.839	0.839	0.849	-	0.780	0.816	0.821	0.754	0.804	0.798	0.795	-
EBO	0.856	0.883	0.963	0.924	0.886	0.896	0.901	-	0.790	0.827	0.773	0.824	0.793	0.798	0.801	-
ReAct	0.852	0.877	0.953	0.900	0.873	0.914	0.895	-	0.786	0.828	0.762	0.822	0.808	0.802	0.801	-
SCALE	0.806	0.839	0.932	0.861	0.835	0.889	0.860	-	0.792	0.826	0.787	0.849	0.813	0.808	0.812	-
Proposed Method																
Dense coverage score integration (ours)	0.891	0.909	0.937	0.921	0.931	0.907	0.916	200	0.670	0.751	0.779	0.763	0.789	0.722	0.746	-

Table 3: OpenOOD results for ImageNet-200. All values reported in AUROC. NFE is the number of function evaluations.

Method	Near-OOD		Far-OOD				Covariate-Shifted ID			NFE
	SSB-hard	NINCO	iNaturalist	Texture	OpenImg-O		ImageNet-C	ImageNet-R	ImageNet-v2	-
Diffusion-Based Methods										
Diffpath 6D	-	-	-	-	-		-	-	-	-
NLL	-	-	-	-	-		-	-	-	-
MSMA	-	-	-	-	-		-	-	-	-
DDPM-OOD	-	-	-	-	-		-	-	-	-
LMD	-	-	-	-	-		-	-	-	-
Dense coverage score integration (ours)	-	-	-	-	-		-	-	-	-
Non-Diffusion Methods										
ODIN	-	-	-	-	-		-	-	-	-
MCD	-	-	-	-	-		-	-	-	-
IC	-	-	-	-	-		-	-	-	-
improved CD	-	-	-	-	-		-	-	-	-

7 DISCUSSION

We discuss how this method can be improve: Find a better integration path, find a stratiedie to select a path. Leverage the reference point issue by using the the log density difference as a norm, and so using other method like knn for OOD detection. Improve diffusion model training in order to have a good approximation of the score even far from the distribution. Quantify the score error made by the diffusion model.

8 CONCLUSION

We introduced a novel density method on diffusion models, showing how to estimate log-likelihood via score integration. Our experiments confirm strong performance on synthetic data and real images embeddings for OOD detection. Future work may explore more sophisticated path choices, better reference-point selection.

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A APPENDIX

A.1 ALGORITHM BASIC

Algorithm 3 Score integration OOD detection

Require: ID training data X_{train} , diffusion model s_θ trained on X_{train} , number of quadrature nodes n , quadrature nodes $(t_i)_{1 \leq i \leq n}$, quadrature weight $(\omega_i)_{1 \leq i \leq n}$, test sample X_{test}

- 1: Initialize $\mu_0 = \text{mean}(X_{train})$
- 2: Refine μ_0 via gradient ascent:
- 3: $\mu_{n+1} \leftarrow \mu_n + \alpha_n s_\theta(\mu_n, 0)$
- 4: $x_{ref} = \mu_N$
- 5: For each $x \in X_{test}$, compute log density difference:
- 6: $f(x) = \frac{1}{2} \sum_{i=1}^n \omega_i s_\theta(z_i, 0) \cdot (x - x_{ref})$ where $z_i = x_{ref} + t_i(x - x_{ref})$
- 7: **return** $(f(x))_{x \in X_{test}}$

A.2 GAUSS-LEGENDRE CONVERGENCE

We can also examine the convergence of the integral as a function of the number of node used in the Gauss-Legendre quadrature. Figure 2 shows that the integral converges rapidly, reaching a stable value for $n \approx 15$.

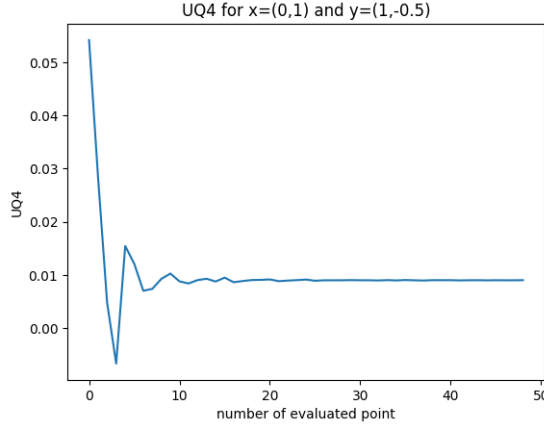


Figure 2: Convergence of the integral as a function of the number of node n

B GAUSS-LEGENDRE ERROR BOUND (KAHANER ET AL.) & DERIVATIVE COMPUTATION

Theorem 1 (Gauss-Legendre Error Bound, adapted from ?). *Let f be $2n$ times continuously differentiable on $[a, b]$, and use the n -point Gauss-Legendre formula*

$$Q_n[f] = \sum_{k=1}^n w_k f(x_k)$$

to approximate $\int_a^b f(x) dx$. Then there exists $\xi \in (a, b)$ such that

$$\int_a^b f(x) dx - Q_n[f] = \frac{(b-a)^{2n+1} (n!)^4}{(2n+1) [(2n)!]^3} f^{(2n)}(\xi).$$

Application to a Straight-Line Path. Consider integrating f_θ along the line segment $\gamma(t) = p_0 + t(p_1 - p_0)$ for $t \in [0, 1]$. Setting $L = \|p_1 - p_0\|$, we define

$$g(t) = f_\theta(\gamma(t)) \|\gamma'(t)\| = f_\theta(\gamma(t)) L.$$

Then

$$\int_{p_0}^{p_1} f_\theta(x) dx = \int_0^1 g(t) dt.$$

Assume g is $2n$ times continuously differentiable on $[0, 1]$. By Theorem 1 (specialized to $a = 0$, $b = 1$), there exists some $\xi \in (0, 1)$ such that

$$\int_0^1 g(t) dt - Q_n[g] = \frac{(n!)^4}{(2n+1) [(2n)!]^3} g^{(2n)}(\xi).$$

Computing the $(2n)$ -th derivative of $g(t)$. Because $\gamma'(t) = p_1 - p_0$ is constant and $g(t) = L f_\theta(\gamma(t))$, repeated application of the chain rule shows

$$g^{(k)}(t) = L [D^k f_\theta(\gamma(t))] (p_1 - p_0, \dots, p_1 - p_0), \quad k \geq 1,$$

where $D^k f_\theta(\gamma(t))$ is the k -th (multilinear) derivative (tensor) of f_θ at the point $\gamma(t)$. Hence, in particular,

$$g^{(2n)}(t) = L [D^{(2n)} f_\theta(\gamma(t))] (p_1 - p_0, \dots, p_1 - p_0).$$

This quantity $g^{(2n)}(t)$ is what appears in the error bound. Thus, to ensure a small quadrature error, one must bound $\max_{t \in [0, 1]} |g^{(2n)}(t)|$ by controlling high-order derivatives of f_θ along γ .